

Information Visualisation

Project Assignment

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Deadlines

1. **Monday, 16th of February at 23:59:** Email to assistant (qi.xu@vub.be) with four group members (Section [2](#))
2. **Sunday, 22nd of February at 23:59:** Canvas upload with top three datasets (Section [3.3](#))
3. **Thursday, 2nd of April:** Interim presentation (Section [7](#))
4. **Thursday, 21th of May:** Final presentation (Section [7](#))
5. **Sunday, 24th of May at 23:59:** Canvas upload with all deliverables (Section [7](#))
6. (Optional) Sunday, 31th of May: Email concerning possible point distributions (Section [2](#))

Note: Deadlines are in Europe/Brussels timezone

1 Goal

The goal of this project is to process and visualise a large dataset using the theoretical knowledge that you will obtain during this course. During the lectures, you will learn various principles, guidelines and techniques that will help you decide on your visualisation. As this is a project counting for 40% of your final grade, you are expected to choose a reasonably sized dataset with an **interactive** visualisation that helps your **target user** to get a good exploration and/or presentation of your data.

2 Groups

Groups are formed with **four** students. The project will be discussed during the first introductory exercise with the deadline for handing in the composition of your group set to the 16th of February at midnight.

You are expected to work in a group towards your final deliverables. Any problems concerning group members should be reported as soon as possible to both the assistant and professor. At the end of the project (before you are graded), you have the possibility to give away a maximum of two points to your other group members. This is usually used in case of unequal task distribution and is completely optional. When sending the mail about this point distribution, make sure that you put all group members in CC. During the oral exam, we will also assess your knowledge of the project.

3 Choosing a Dataset

We expect the groups to use a dataset-oriented approach towards their final visualisation. This means that you should first find a dataset that you like and then define the tasks needed to gradually process and visualise this dataset.

Datasets can be taken from anywhere but **have to be approved by us**, meaning you have to choose an existing dataset that is big enough to be a challenge. We advise that you decide on a **topic** first, then find three potential datasets about that topic. Once you have your main dataset you will have to decide the **target user** and tasks that you are going to create a visualisation for. This is important as you will have to motivate why this user needs this visualisation and evaluate your final result with those users.

3.1 Sources

The following list is a helpful set of sources where you can find public datasets. These sources, including a few more domain-specific ones, will be discussed during exercise 1:

- Kaggle: <https://www.kaggle.com/datasets>
- Zenodo: <https://zenodo.org>
- Google Dataset Search: <https://datasetsearch.research.google.com>
- DataMed: <https://datamed.org>

Of course, you can pick any source you want as long as the dataset is publicly available.

3.2 Requirements

The following list contains some soft requirements set for the dataset. This list helps you decide on the three datasets and our method of approving the datasets:

- **Size:** You can combine multiple datasets together, but your main dataset should be reasonably sized (both in the amount of data that is available and the information that is represented).
- **Real data:** The data that your dataset contains should be real data. Generated data that is not based on any real source is not allowed (e.g. randomly generated sales).
- **Meta-information:** Related to the “real data”, the dataset that you use should contain information about how the information was obtained. On the dataset sources that we will discuss, this is usually the case.

3.3 Approval

Once you have selected **three** potential datasets, you have to upload a description to canvas (does not have to be a report, can be a text file). For each of the three datasets you should provide:

- The source of the dataset (where can we download it)
- In case the above link does not contain meta-information on how the data was retrieved, make sure that you specifically mention it
- A short summary of the dataset
- Target users (who are you going to create a visualisation for)
Note: It is important for your final evaluations that you design the project for a specific users
- Potential initial ideas you have (e.g. do you want to combine or compare data, do you already have an initial idea to visualise it ...)

We will respond as soon as possible with our feedback on each dataset.

4 Preprocessing the Dataset

Preprocessing a dataset is part of the assignment and inevitable with any dataset. This processing could be to combine multiple datasets or to rearrange the data before creating visualisations. You are not required to perform this preprocessing in the same framework that you use for the visualisation. While many frameworks offer this capability, it is often easier to choose a specific framework depending on the type of data that you are dealing with. During the exercises, you will learn dataset preprocessing using Python.

Note: The main goal of the project is to visualise information. While the preprocessing is graded, it should not be the main aspect of your project.

5 Visualising the Dataset

Once you have chosen a dataset and prepared it, you can start to visualise it using any framework that is capable of doing so. Keep in mind that *creativity is awarded*.

It is of utmost importance that you correctly apply the theoretical knowledge and guidelines from the lectures, including the use of interactive visualisations.

A small list of frameworks:

- R (Studio): Used by academics for stationary visualisation and preprocessing. The Shiny¹ package can be used to create interactive web interfaces.
- D3.js: Commonly used visualisation framework on the web, can be used on top of MapBox or Leaflet
- Plot.ly: Visualisation framework that works for R, D3.js and Python
- Leaflet.JS: Useful for geospatial visualisation
- MapBox: Similar to leaflet
- Python: Commonly used visualisation framework by academics and programmers. You can use frameworks such as Bokeh for interactive visualisations.
- p5.js: Creative coding framework widely used for browser-based visualisation interactions

Note that business intelligence tools (e.g. Tableau or Power BI) where no programming is necessary, are not allowed.

6 Evaluating the Visualisation

The validation and evaluation of your solution is an important aspect of any academic project, including your visualisation. Information about this evaluation will be discussed in the lectures and exercises.

¹<https://shiny.rstudio.com>

7 Presentations and Final Report

During this course, you will have to perform two presentations. Both presentations will be held **on campus**. Timeslots for these presentations will be published as an announcement on Canvas. Every team member is expected to participate in the **live** presentation

The first presentation is an interim progress presentation on the **2nd of April**. During this progress presentation you have **10 minutes** to present your chosen dataset, the steps you have taken towards processing and visualising the dataset and a GANTT chart of your future progress. Make sure to mention why you have chosen the dataset and for which task(s) you are creating a visualisation and how you are bringing the data to the target user. Immediately following the presentation you will receive feedback on your progress and the provided GANTT chart.

The final presentation on **May 21** should be a complete overview of your project. Similar to the first presentation, you discuss your dataset, the preprocessing performed on these datasets, the visualisation(s) you have made and the evaluations. For this presentation, you have **15 minutes** with an additional 5 minutes for questions. After the presentation, you will receive potential feedback on things that are important to outline in your final report (such as aspects that were not clear from your presentation).

The deadline for the final report and visualisation deliverables (dataset, source code of preprocessing and visualisation) is on **Sunday the 24th of May at midnight** by uploading it to Canvas. If your deliverables are too large to upload to Canvas, you can also upload them to the SharePoint of the VUB. Make sure that you provide a report with the scientific quality expected from a Master course (e.g. use of references, VUB layout, writing style, ...). The report should have a minimum of 10 pages (with images but without possible appendices) describing your complete solution **including setup instructions on how we can run and view your result**.

8 Referencing

Like any academic project, all deliverables, including your report, images and source code are checked for plagiarism. Make sure that you always reference sources in both your source code, report and **your final visualisation**. This also includes the dataset(s) that you have chosen.

9 Deliverables

Your final deliverables should consist of:

- Interactive visualisation
 - Demo video showcasing the functionalities
 - Source code of the visualisation
 - Source code of the preprocessing
 - (optional) Website URL of the live version
- Interim and final presentations
- Evaluation of your solution with target users
- A report of your visualisation, evaluation and preprocessing

10 Examples

Throughout the lectures, you will see various real-world examples. The following list shows several examples that represent the quality that we are after:

- [Coronavirus 2019](#)
Provides a useful dashboard of information about infections, deaths and recoveries. Also notice how the visualisation is accompanied by additional information containing the sources. Note that this visualisation also has several mistakes, such as its slow loading time.
- [Codex Atlanticus](#)
An interactive visualisation that makes it easy to see what pages of the Codex Atlanticus contain certain information. The visualisation is good in clearly visualising how much information is included on each page.
- [SelfieCity](#)
This interactive visualisation proves that you do not need textual or relational datasets. On this site you can see what type of selfies are taken around the world including interactive comments that show that certain countries have more smiles on selfies than others for example. The site also contains some findings based on their datasets that are also visually represented.

11 Grading

This section gives you an indication on how the grading of the project is performed. Each project is unique in a way that different areas have different priorities for the project. For instance, a project that is very CPU intensive will have different challenges on the pre-processing side to support the required

interactions etc. In general, the following matrix will be applied to determine your grade.

11.1 Creativity & Interactivity (10%)

The project should support a broad set of interactions that empower the target user to perform their tasks in an efficient way. Creativity will be rewarded.

11.2 Design & Guidelines (30%)

The project design is an important factor of the project. First and foremost, the project should be well planned beforehand with preliminary research. Based on this research, the student should come up with a possible solution that can aid their selected target user to achieve the target goal. The final design should be complete and allow the end user to perform all actions necessary to achieve their goals. It is of utmost importance that the created visualisation adheres to the set of design guidelines discussed during the lectures.

11.3 Frameworks (10%)

Throughout the course we will discuss and experiment with multiple visualisation frameworks. Students are expected to select a framework that fits their target visualisation. Furthermore, students who demonstrate an understanding of the concepts of their chosen framework and who implement their solution using the strengths of the framework will be rewarded.

11.4 Quality of the Report + Extras (5%)

General report grading based on the scientific quality, references, completeness, correctness and structure. This grading also includes any extra artefacts such as a demo video.

11.5 Dataset & Target User (20%)

The selection of the dataset and the target goal is the first step of the project. The student should evaluate their datasets based on the criteria discussed in the lectures as well as during the second exercise session. It is also very important that the selection of the dataset happens with the specific end user and goal in mind, I.e. certain applications might have different requirements in terms of accuracy and completeness of the data than others.

11.6 Code (10%)

Projects are expected to follow industry best practises in terms of separation of concerns, architecture and general design. Projects that structure their code to take future change requests into account will be rewarded.

11.7 Presentation (5%)

General presentation grading based on the slideware, demonstration, timing, content and structure.

11.8 Evaluation (10%)

The project should contain an extensive evaluation on the design(s). Grading is performed based on the soundness of the evaluations, the scale of the evaluation and its richness. Correct design practices should be applied (these will also be discussed during the exercise sessions). The students should be able to draw conclusions from the evaluations to aid in writing a future work and conclusion on the project.